Solve fraction addition problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________ Date: ______________

1. Read the model: A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug? Copy or complete the first step.

2. Calculate 3/4 + 5/6.

3. Calculate 7/8 - 5/12.

4. Miro says: Miro adds or subtracts both the numerators and denominators. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug? Copy or complete the first step.
   Answer 2 5/12 litres.

2. Calculate 3/4 + 5/6.
   Answer 1 7/12.

3. Calculate 7/8 - 5/12.
   Answer 11/24.

4. Miro says: Miro adds or subtracts both the numerators and denominators. Is this correct? Explain.
   Answer Use equivalent fractions with a common denominator, then operate on the numerators only.

Solve fraction addition problems

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ___________________

1. A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug?

2. Calculate 3/4 + 5/6.

3. Calculate 7/8 - 5/12.

4. Solve this independently and show every stage: A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug?

5. Use a second method or representation for: Calculate 3/4 + 5/6.

6. Check this result using an inverse, estimate or known fact: Calculate 7/8 - 5/12.

7. Prove or explain your reasoning: Calculate 2 1/3 + 1 5/6.

8. Identify and correct this misconception: Miro adds or subtracts both the numerators and denominators.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Calculate 7/8 - 5/12.

Teacher answers and checking prompts

1. A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug?
   Answer 2 5/12 litres.

2. Calculate 3/4 + 5/6.
   Answer 1 7/12.

3. Calculate 7/8 - 5/12.
   Answer 11/24.

4. Solve this independently and show every stage: A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug?
   Answer 2 5/12 litres.

5. Use a second method or representation for: Calculate 3/4 + 5/6.
   Answer Expected result: 1 7/12. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Calculate 7/8 - 5/12.
   Answer Expected result: 11/24. A valid check must be shown.

7. Prove or explain your reasoning: Calculate 2 1/3 + 1 5/6.
   Answer 4 1/6. The explanation must show why the method works.

8. Identify and correct this misconception: Miro adds or subtracts both the numerators and denominators.
   Answer Use equivalent fractions with a common denominator, then operate on the numerators only.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Calculate 7/8 - 5/12.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve fraction addition problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ______________

1. Prove or explain your reasoning: Calculate 2 1/3 + 1 5/6.

2. Prove the answer to this question, rather than only calculating it: A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug?

3. Solve this in two different ways and compare them: Calculate 3/4 + 5/6.

4. Work backwards from the answer to reconstruct the question: 11/24.

5. Explain why this reasoning fails: Miro adds or subtracts both the numerators and denominators.

6. Create a new example that tests the same idea as: Calculate 7/8 - 5/12.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate 2 1/3 + 1 5/6.
   Answer 4 1/6. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A jug contains 3 1/2 litres. 1 3/4 litres are used and 2/3 litre is added. How much is in the jug?
   Answer Expected result: 2 5/12 litres. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Calculate 3/4 + 5/6.
   Answer Expected result: 1 7/12. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 11/24.
   Answer One valid reconstruction is: Calculate 7/8 - 5/12. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro adds or subtracts both the numerators and denominators.
   Answer Use equivalent fractions with a common denominator, then operate on the numerators only.

6. Create a new example that tests the same idea as: Calculate 7/8 - 5/12.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.